

Smart Contract Audit Report

Generated: 2026-07-08 18:26:04

Total Vulnerabilities	Risk Score	Slither Findings	AI Findings
6	8/10	3	3

Overall Assessment

The contract has several security vulnerabilities, including reentrancy and lack of access control, which can be exploited by attackers. Additionally, the contract lacks input validation and event emission, which can lead to unexpected behavior and make it difficult to track activity.

Slither Static Analysis Findings

Detector	reentrancy-eth
Severity	High
Confidence	Medium
Description	Reentrancy in Vulnerable.withdraw() (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#21-31) ■ External calls: ■ - (success,None) = msg.sender.call{value: amount}() (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#21-31) ■ State variables written after the call(s): ■ - balances[msg.sender] = 0 (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#29) ■ Vulnerable.balances (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#9) can be used in cross-contract calls: ■ - Vulnerable.balances (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#9) ■ - Vulnerable.deposit() (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#13-17) ■ - Vulnerable.withdraw() (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#21-31)

Detector	solc-version
Severity	Informational
Confidence	High
Description	Version constraint ^0.8.0 contains known severe issues (https://solidity.readthedocs.io/en/latest/bugs) ■ - FullInlinerNonExpressionSplitArgumentEvaluationOrder ■ - MissingSideEffectsOnSelectorAccess ■ - AbiReencodingHeadOverflowWithStaticArrayCleanup ■ - DirtyByteArrayToStorage ■ - DataLocationChangeInInternalOverride ■ - NestedCalldataArrayAbiReencodingSizeValidation ■ - SignedImmutables ■ - ABIDecodeTwoDimensionalArrayMemory ■ - KeccakCaching. It is used by: ■ - ^0.8.0 (C:/Users/abhap/AppData/Local/Temp/tmpcu1hak7n.sol#3)

Detector	low-level-calls
Severity	Informational
Confidence	High
Description	Low level call in Vulnerable.withdraw() (C:/Users/abhapp/AppData/Local/Temp/tmpcu1hak7n.sol#21-32) - (success,None) = msg.sender.call{value: amount}() (C:/Users/abhapp/AppData/Local/Temp/tmpcu1hak7n.sol#21-32)

AI Analysis Findings

Name	Unprotected Function
Severity	Medium
Description	The deposit and withdraw functions are not protected against unauthorized access, allowing anyone to interact with the contract.
Fix	Add access control mechanisms, such as onlyOwner or role-based access control, to restrict access to sensitive functions.
Fixed Code	function deposit() public onlyOwner payable { balances[msg.sender] += msg.value; }

Name	Lack of Input Validation
Severity	Low
Description	The contract does not validate user input, which can lead to unexpected behavior or errors.
Fix	Add input validation to ensure that user input is valid and within expected ranges.
Fixed Code	function deposit() public payable { require(msg.value > 0, 'Invalid deposit amount'); balances[msg.sender] += msg.value; }

Name	No Event Emission
Severity	Low
Description	The contract does not emit events for important state changes, making it difficult to track activity.
Fix	Add event emission to notify users of important state changes.
Fixed Code	event Deposit(address indexed sender, uint256 amount); function deposit() public payable { balances[msg.sender] += msg.value; emit Deposit(msg.sender, msg.value); }